

FREEDOM INTERNATIONAL SCHOOL

WORKSHEET

PHYSICS

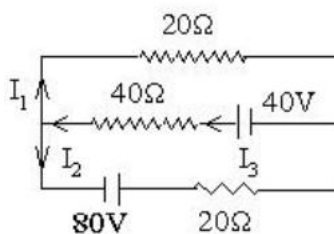
CLASS XII

CURRENT ELECTRICITY

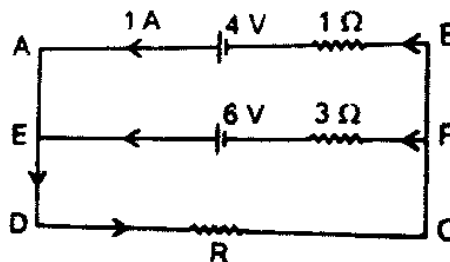
1. A 10 V battery of negligible internal resistance is connected across a 200 V battery and a resistance of $38\ \Omega$ as shown in the figure. Find the value of current in the circuit.



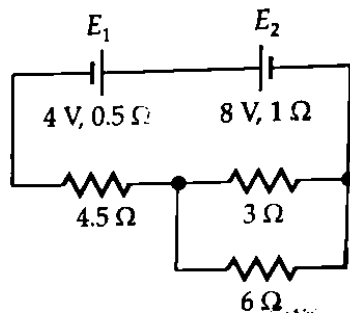
2. A voltmeter of resistance $998\ \Omega$ is connected across a cell of emf 2 V and internal resistance $2\ \Omega$. Find the p.d across the voltmeter, that is connected across the terminals of the cell.
3. The reading on a high resistance voltmeter when a cell is connected across it is 2.2 V. When the terminals of the cell are also connected to a resistance of $5\ \Omega$, the voltmeter reading drops to 1.8 V. Find the internal resistance of the cell.
4. Two identical cells of emf 1.5 V each joined in parallel provide supply to an external circuit consisting of two resistances of $17\ \Omega$ each joined in parallel. A very high resistance voltmeter reads the terminal voltage of the cells to be 1.4 V. Calculate the internal resistance of each cell.
5. Using Kirchhoff's rules, determine the value of the current I_1 in the electric circuit given below.



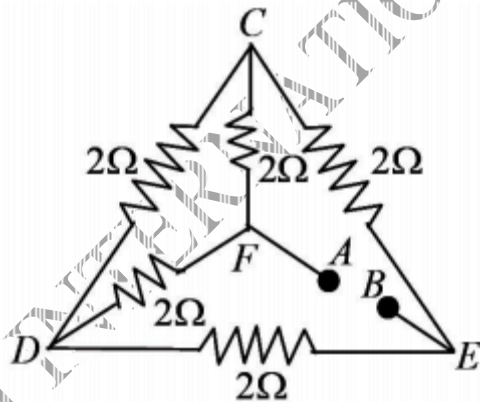
6. Using Kirchhoff's rules, determine (i) the voltage drop across the unknown resistor R and (ii) the current flowing in the arm EF in the circuit as shown.



7. In the circuit diagram given in the figure, the cells E_1 and E_2 have emfs 4 V and 8 V and internal resistances $0.5\ \Omega$ and $1.0\ \Omega$ respectively. Calculate the current in each resistance.



8. A series battery of 10 lead accumulators, each of emf 2 V and internal resistance $0.25\ \Omega$, is charged by a 220 V d.c mains. To limit the charging current, a resistance of $47.5\ \Omega$ is used in series in the charging circuit. What is (a) the power supplied by the mains and (b) power dissipated as heat? Account for the difference of power in (a) and (b).
9. A potential difference of 2 V is applied between the points A and B shown in network drawn in the figure. Calculate (i) the equivalent resistance of the network between the points A and B, and (ii) the magnitudes of currents flowing in the arms AFCEB and AFDEB.



10. In the given circuit, with steady current, calculate the potential difference across the capacitor and the charge stored in it.

